

Notes: Petersen and Rajan (QJE, 1995)

The Effect of Credit Market Competition on Lending Relationships

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1 Big picture

- **Question.** Does stronger competition in credit markets make lending relationships less valuable, especially for young or opaque firms that need financing before their quality is well understood?
- **Setting.** The paper studies small U.S. firms using the National Survey of Small Business Finances and exploits cross-sectional variation in local banking-market concentration, measured by the deposit-market Herfindahl index in the firm's county or Metropolitan Statistical Area.
- **Core challenge.** Relationship lending often requires **intertemporal sharing of surplus**: lenders may need to subsidize a firm early and recover those rents later. In competitive credit markets, banks may be unable to do this because successful borrowers can refinance elsewhere.
- **Main theoretical result.** When lenders have market power, they can backload repayment over time. This makes them willing to finance **lower-quality, young, or distressed firms** that would be rationed in competitive markets.
- **Main empirical result.** Young firms are more likely to obtain institutional finance in concentrated credit markets than in competitive ones. These differences do not appear to be driven by better firm quality in concentrated markets. Moreover, firms in concentrated markets appear less credit constrained according to their trade-credit behavior.
- **Interest-rate result.** Loan rates for young firms start lower in concentrated markets, but they fall much more slowly with age than in competitive markets. This is exactly the intertemporal smoothing pattern suggested by the model: banks subsidize young firms and recover rents later.
- **Why the paper matters.** The paper gives one of the classic arguments that more competition is not always better in credit markets. There is a trade-off between **credit availability for young firms** and **the long-run price of credit for successful survivors**.

2 Data and measurement (key objects)

2.1 Observed firm-level sample

- **Data source.** The data come from the **National Survey of Small Business Finances** (NSSBF), conducted in 1988–1989 by the Federal Reserve Board and the U.S. Small Business Administration.
- **Target population.** The survey covers **nonfinancial, nonfarm** small businesses operating as of December 1987.
- **Firm size restriction.** All firms in the survey have fewer than 500 employees.
- **Sample size.** There are 3,404 firms in the sample, of which 1,875 are corporations (including S corporations) and 1,529 are partnerships or sole proprietorships.
- **Basic firm moments.** Mean book assets are about \$1.05 million, the median is about \$130,000, mean sales are about \$2.6 million, median sales are about \$300,000, and the median firm age is 10 years.
- **Other firm characteristics.** About 90% of the firms are owner-managed. The sample is stratified by census region, urban versus rural location, and employment size.

2.2 Observed borrowing, financing, and relationship measures

- **Institutional debt.** The paper studies whether a firm borrows from institutional lenders and how much it borrows relative to assets.
- **Composition of debt.** Average debt-to-asset ratio is about 0.33. Roughly 80% of debt comes from institutions; within institutional debt, about 80% is from banks and about 15% from nonbank financial institutions.
- **Trade credit.** Trade credit financing is roughly 10% of assets, or about one-third as large as debt. The paper uses the percentage of early-payment discounts taken and days payable outstanding (DPO) as indicators of how constrained a firm is in institutional credit markets.
- **Relationship variables.** The survey reports the **longest relationship** with a financial institution, the number of institutions from which the firm borrows, and whether the institutions providing debt also provide significant financial services.
- **Loan-rate subsample.** For a subset of 1,277 firms, the survey reports the interest rate on the firm's most recent loan. This is used to test whether the life-cycle path of borrowing costs differs by market concentration.

2.3 Local credit-market structure

- **Competition proxy.** The key variable is the deposit-market Herfindahl index of the county or Metropolitan Statistical Area where the firm is headquartered.
- **Three market categories.**
 - most competitive: Herfindahl < 0.10 ,
 - middle market: Herfindahl between 0.10 and 0.18,
 - most concentrated: Herfindahl > 0.18 .
- **Why this is plausible.** Small-business borrowing is local. More than half the firms are within 2 miles of their primary financial institution, and 90% are within 15 miles.
- **Additional validation.** Solicitation by lenders is more frequent in competitive markets, especially for older firms, which is consistent with the Herfindahl index capturing meaningful variation in lender competition.

2.4 Unobservables and timing

- **Unobserved firm type.** In the model, lenders do not initially know whether a borrower is “good” or “bad,” but learn a lot after a period of interaction.
- **Cross-sectional age as life cycle.** The empirical analysis is cross-sectional, so the paper uses firm age to infer how access to credit and loan pricing vary over the firm's life cycle.
- **Main concern.** Older firms may differ from young firms because of survivorship, not just because lending relationships evolve over time. The paper therefore spends substantial effort arguing that differential survival across markets is unlikely to explain the results.

3 Models and empirical specifications (all equations + interpretation)

3.1 Environment and production opportunities

There are two types of entrepreneurs: **good** and **bad**. At date 0, a good entrepreneur can choose either a safe or a risky project.

- **Safe project at date 0.** Investment I_0 yields payoff S_1 at date 1. If the project succeeds, the entrepreneur can invest I_{1S} in another safe project that yields S_2 at date 2.
- **Risky project at date 0.** Investment I_0 yields R_1 with probability p and 0 otherwise. If it succeeds, the entrepreneur can invest I_{1R} in another project yielding R_2 at date 2.
- **Bad entrepreneurs.** Their projects always fail and return zero at date 1.

The paper imposes the assumptions

$$\begin{aligned} S_2 + S_1 - I_{1S} - I_0 &> 0, \\ p(R_2 + R_1 - I_{1R}) - I_0 &< 0, \\ pR_2 = S_2 &> pI_{1R} = I_{1S}, \\ I_{1S} &> R_1 > S_1, \end{aligned}$$

and

$$\frac{S_2}{I_{1S}} > M \geq 1.$$

- **Interpretation.** The safe project is socially valuable; the risky project has negative expected net present value but can be privately attractive if repayment is structured badly.
- **Role of M .** M measures the lender's market power. When $M = 1$, the credit market is competitive. When $M > 1$, the lender can earn rents in future periods.

3.2 Finance, information, and debt contracts

At date 0, banks know only that a fraction θ of entrepreneurs are good. At date 1, after dealing with the borrower, the bank becomes fully informed about the entrepreneur's type.

- **Debt only.** Banks can hold only debt claims, not equity claims.
- **No state-contingent contract on project choice or type.** The bank cannot write a contract contingent on whether the entrepreneur chose the safe or risky project, or on whether the entrepreneur is good or bad.
- **Economic meaning.** This is what creates the classic adverse-selection and moral-hazard problem: high initial repayments may induce good entrepreneurs to gamble.

3.3 The entrepreneur's incentive constraint

A good entrepreneur borrows I_0 at date 0 and promises repayment D_1 at date 1.

If he takes the safe project, then at date 1 he must borrow

$$I_{1S} - (S_1 - D_1). \quad (1)$$

His expected continuation profit is

$$\max [S_2 - M(I_{1S} - (S_1 - D_1)), 0]. \quad (2)$$

If instead he takes the risky project, expected profit is

$$\max \{p [R_2 - M(I_{1R} - (R_1 - D_1))], 0\}. \quad (3)$$

Using the assumptions above, the good entrepreneur prefers the safe project if

$$\frac{S_1 - pR_1}{1 - p} \geq D_1. \quad (4)$$

- **Interpretation.** D_1 cannot be too high. If the initial promised repayment is too large, the entrepreneur shifts into the risky project to try to make the payment.
- **Economic intuition.** This is the key distortion: front-loading repayment strengthens moral hazard.

3.4 The bank's participation condition and the financing cutoff

For the bank to lend at date 0, it must recover its initial investment while accounting for the rents it can extract at date 1. The lending condition is

$$D_1 \geq \frac{I_0}{\theta M} - \frac{M - 1}{M}(I_{1S} - S_1). \quad (5)$$

Combining the entrepreneur's incentive constraint with the bank's participation constraint, only entrepreneurs with ex ante credit quality above the cutoff

$$\theta^c(M) = \frac{I_0(1 - p)}{M(S_1 - pR_1) + (M - 1)(I_{1S} - S_1)(1 - p)} \quad (6)$$

get financed.

- **Result 1.** As lender market power M rises, the quality threshold $\theta^c(M)$ falls. Therefore, **lower-quality firms obtain finance in more concentrated credit markets.**
- **Why.** Greater market power lets the lender recover some rents later, so it can accept lower repayments initially and still break even.

3.5 Intertemporal smoothing of credit prices

At date 1, the lender charges the gross rate M . At date 0, however, the face value D_1 is bounded by

$$\min \left[\frac{I_0 M}{\theta}, \frac{S_1 - p R_1}{1 - p} \right] \geq D_1 \geq \frac{I_0}{\theta M} - \frac{M - 1}{M} (I_{1S} - S_1). \quad (7)$$

Let t be the ratio of the face value charged per dollar lent at date 1 to the face value charged per dollar lent at date 0. Then, for banks with market power $M > M'$,

$$t(M) \geq t(M'). \quad (8)$$

If θ has positive density on $[0, 1]$, then

$$\text{Average}(t(M)) \geq \text{Average}(t(M')). \quad (9)$$

- **Result 2.** The lowest-quality young firm financed by a lender with more market power pays a **lower initial interest rate** than if financed by a lender in a more competitive market.
- **Result 3.** The relative decline in borrowing costs over the firm's life cycle is **smaller** when lender market power is greater.
- **Interpretation.** Concentrated markets permit **intertemporal interest-rate smoothing**: low rates when the firm is young, high rates when it is established.

3.6 Why standard contracts do not trivially solve the problem

- **Debt versus equity.** The paper assumes banks cannot hold equity, but also argues that even if equity were feasible, explicit contracts may be hard to enforce or unattractive for regulatory and political reasons.
- **Long-term debt is not enough.** A long-term fixed-payment debt contract cannot make repayment vary with the project actually taken. Market power, by contrast, lets the lender extract more from the borrower precisely when the borrower's project succeeds.
- **Bonding contracts.** Clauses such as a first-right-of-refusal or commitments to return to the same bank could, in principle, mimic some of the backloading. The paper argues that such contracts may be costly, incomplete, or hard to enforce in practice.

3.7 Empirical implications from the model

The paper restates the theory in three empirical predictions:

- Provided the quality distribution is similar across markets, relatively more firms should obtain institutional credit in more concentrated markets, and the average quality of financed firms should be lower there.
- Credit should be cheaper for the lowest-quality firms in concentrated markets.
- Borrowing costs should fall faster with age in competitive markets than in concentrated markets.

3.8 Institutional debt and credit availability in the data

The first empirical exercise is descriptive: compare the fraction of firms with institutional debt, and the institutional debt-to-asset ratio, across market-concentration categories and age bins.

- **Interpretation.** This is the most direct test of Result 1.
- **Caveat.** For older firms, actual borrowing may reflect both demand and supply, so the paper also uses an alternative measure of credit availability.

3.9 Trade-credit discounts as a measure of binding credit constraints

The paper argues that firms that cannot get enough cheap institutional finance will borrow at the margin from expensive noninstitutional sources. Trade credit is a natural candidate.

The key variable is the percentage of offered early-payment discounts that the firm actually takes. Let the latent desired fraction of discounts taken be

$$D_i^* = X_i' \beta + \varepsilon_i,$$

but the observed fraction is censored between 0 and 100 percent:

$$D_i = \max\{0, \min(D_i^*, 100)\}.$$

The paper therefore estimates a **two-sided Tobit**.

A reduced-form representation of the estimating equation is

$$D_i^* = \beta_0 + \beta_1 \log(\text{Assets}_i) + \beta_2 \frac{\text{Operating Profits}_i}{\text{Assets}_i} + \beta_3 \mathbf{1}\{\text{corporation}\} + \beta_4 \log(\text{Age}_i) + \beta_5 \text{Relationship}_i + \beta_6 \text{Competition}_i$$

where X_i includes urban-rural, region, industry, sales growth, and state banking-law controls.

- **Interpretation.** A higher percentage of discounts taken means the firm is **less credit constrained**.
- **Why this works.** Early-payment discounts are economically very large and are argued to be largely industry-specific rather than customized firm by firm.
- **Two critical conditions.** The marginal cost of trade credit must exceed the marginal cost of institutional credit, and trade-credit terms must be roughly comparable for similar firms.

3.10 Loan-rate specification

For the interest-rate test, the paper estimates a regression in which the effect of age differs across market structures:

$$r_i = \alpha_c + \beta_c \log(\text{Age}_i) + \gamma Z_i + u_i,$$

where c indexes the concentration category.

The control vector Z_i includes:

- prime rate,

- term-structure spread,
- default spread,
- floating-rate indicator,
- collateral dummies,
- lender-type dummies,
- firm size, leverage, and corporate form,
- relationship variables,
- industry and region controls.
- **Interpretation.** The intercept captures how costly credit is for the newest firms in each market structure. The slope on log age captures how quickly credit gets cheaper as the firm survives and becomes better known.
- **Prediction.** Competitive markets should have a steeper negative age slope.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in a naive interpretation

A raw comparison of borrowing across concentrated and competitive credit markets could be misleading if:

- firms are inherently better in concentrated markets,
- rural areas have better soft information and thus easier credit,
- regions with concentrated banking happen to avoid negative shocks,
- older firms in concentrated markets survive at different rates, changing the composition of the sample.

4.2 Why local deposit concentration is a reasonable competition proxy

- Small-business credit markets are local: over half the firms are within 2 miles of their primary institution and 90% within 15 miles.
- Solicitation of business is also local, and solicitation rates are higher in competitive markets.
- The Herfindahl index is therefore not just a banking-market statistic; it plausibly captures the ease with which a firm can shop for alternative lenders.

4.3 Why firm-quality differences are unlikely to explain the main results

- The firms in competitive markets are not better only in financing outcomes; if anything, Table I shows they are as good as or better than firms in concentrated markets on observable profitability measures.

- Industry composition is similar across concentrated and competitive markets.
- The regional correlation between Herfindahl concentration and broad census-region dummies is small.

4.4 Why trade-credit behavior is informative about rationing

- Firms offered trade credit can substitute into it when institutional credit is insufficient.
- The paper argues that the cost of passing up early-payment discounts is extremely high, making it a credible measure of expensive marginal borrowing.
- Since discount terms are largely standardized within industries, differences in discounts taken are more plausibly about financing constraints than about idiosyncratic contract design.

4.5 Why the interest-rate life-cycle profile is informative

- A pure “small-town information” story would imply uniformly lower rates in concentrated markets, not rates that are lower only for young firms and higher for old firms.
- The model predicts a specific **cross-over pattern**: concentrated markets subsidize the young and recoup from the old.
- The observed life-cycle pattern of rates is therefore much closer to intertemporal smoothing than to a one-shot information or regional-shock story.

4.6 Robustness checks

The paper checks the main conclusions against:

- urban versus rural controls (MSA indicator),
- unit banking state controls,
- sales-growth controls,
- detailed industry controls,
- survivorship arguments based on observable firm characteristics.

The central results remain intact.

4.7 Relation to the literature

- **Theory link.** The paper draws on Diamond (1989) and Stiglitz and Weiss (1981): adverse selection and moral hazard can generate rationing.
- **Relationship lending.** It complements the relationship-lending literature by showing that the value of a relationship depends on the surrounding degree of competition.
- **Broader message.** Competition may weaken relational contracts not only in labor markets but also in credit markets.

5 Tables and figures

5.1 Table I and Figure I: Are firms in concentrated markets different?

TABLE I
SUMMARY STATISTICS FOR FIRMS CLASSIFIED BY AGE AND
CREDIT MARKET CONCENTRATION

Firm characteristics	Young firms			Old firms		
	Most competitive	Middle market	Most concentrated	Most competitive	Middle market	Most concentrated
Firm solicited by financial institution during the year (1 = yes)	0.32 ¹⁰ (0.46) 166	0.32 (0.47) 488	0.26 (0.44) 966	0.46 ¹ (0.50) 112	0.39 (0.49) 516	0.31 (0.46) 972
Firm size (in \$1000s)	863	1074	569	1296	1692	1171
Book value of assets	102 (3247) 179	108 (7188) 517	103 (1836) 1010	151 (2722) 117	207 (5379) 554	155 (3062) 1027
Gross profits	2.78 ¹	2.18	1.88	2.39 ¹	1.95	1.79
Assets	1.61 ¹ (2.96) 170	1.24 (2.46) 491	1.06 (2.21) 965	1.36 ¹ (2.76) 114	1.66 (2.31) 520	0.97 (2.21) 973
Operating profits	1.14	1.05	0.63	0.61	0.62	0.64
Assets	0.17 (4.84) 170	0.18 (5.74) 491	0.15 (2.49) 965	0.23 ¹⁰ (1.19) 114	0.14 (1.93) 520	0.13 (2.88) 973
Sales growth (1986–1987)	0.30 (0.98) 140	0.32 (0.92) 435	0.31 (0.91) 843	0.15 (0.63) 109	0.15 (0.68) 495	0.14 (0.58) 934
Number of instits. from which firm borrows	1.35 1.00 (0.59) 99	1.40 1.00 (0.62) 291	1.37 1.00 (0.62) 655	1.28 ¹ 1.00 (0.51) 71	1.36 1.00 (0.65) 323	1.43 1.00 (0.69) 624
Longest business relation with financial institution (in years)	4.10 ¹ 4.00 (3.09) 179	4.66 4.00 (2.93) 517	4.75 4.00 (2.95) 1010	16.37 15.00 (9.81) 117	16.42 15.00 (10.60) 554	17.42 15.00 (11.05) 1027

The SBA survey classifies the Metropolitan Statistical Area or county where the firm is headquartered into three categories on the basis of the commercial bank deposit Herfindahl index of the area. The most competitive markets are those with a value of the Herfindahl index of less than 0.10. The most concentrated markets are those with a Herfindahl index of more than 0.18. The middle market is those markets where the Herfindahl index is between 0.10 and 0.18. Young firms are ten years old (the median age) or less. Old firms are more than ten years old. We report the mean, the median, the standard deviation, and the number of observations in each cell.

For each variable we tested the equality of the means in the most concentrated and most competitive markets. We tested the equality of the medians in the most concentrated and most competitive markets using the Wilcoxon Rank-Sum test. Both tests were done separately for young and old firms. Differences in the means and median is noted for significance levels of 10 percent (¹⁰), 5 percent (⁵), and 1 percent (¹).

Distribution of Industries
by Deposit Market Concentration

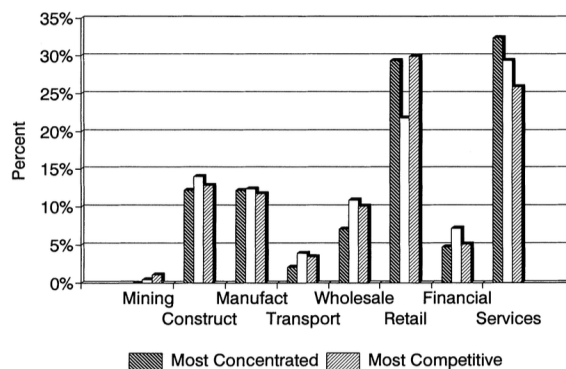


FIGURE I

Firms are classified by their one-digit SIC code. We graphed the percent of firms in each industry for each of the deposit market concentrations: most competitive (Herfindahl index less than 0.10), most concentrated (Herfindahl index greater than 0.18), and the middle market.

Read:

- The basic sample does *not* suggest that firms in concentrated markets are mechanically better firms. Among young firms, median book assets are essentially identical: about \$102,000 in competitive markets and \$103,000 in concentrated markets.
- Young firms in competitive markets actually look at least as profitable on several observable dimensions, which biases against finding easier credit in concentrated markets.
- Industry composition is also broadly similar across market structures, as shown in Figure I.
- Table I also provides a useful validation of the competition proxy: older firms are solicited much more often in competitive markets (46%) than in concentrated markets (31%), which is exactly what one would expect if concentrated markets are less competitive.

5.2 Table II: Institutional indebtedness over the firm life cycle

Read:

- Young firms are much more likely to obtain institutional debt in concentrated markets. For firms

age 10 or below, 65% borrow institutionally in the most concentrated markets versus 55% in the most competitive markets.

- The result is strongest for the youngest firms. Among firms age 4 or less, only 48% have institutional debt in the most competitive markets, compared with 65% in the most concentrated markets.
- The debt-to-asset ratio of young institutional borrowers is also slightly higher in concentrated markets, which fits the idea that young firms face fewer financing constraints there.
- For older firms the difference in access disappears, and the pattern in debt usage partly reverses, which is consistent with banks recovering early subsidies later in the relationship.

TABLE II
INSTITUTIONAL INDEBTEDNESS OF SMALL FIRMS

Firm age (in years)	Most competitive market		Middle market		Most concentrated market	
	Firms with instit. debt	Debt assets	Firms with instit. debt	Debt assets	Firms with instit. debt	Debt assets
	Percent number	Mean std. dev.	Percent number	Mean std. dev.	Percent number	Mean std. dev.
≤2	0.47	0.49	0.49	0.44	0.65	0.46
	12	(0.36)	55	(0.46)	147	(0.38)
3–4	0.49	0.44	0.60	0.39	0.65	0.43
	20	(0.35)	60	(0.49)	141	(0.30)
5–6	0.65	0.28	0.55	0.43	0.68	0.39
	17	(0.26)	64	(0.56)	131	(0.36)
7–8	0.69	0.19	0.63	0.38	0.64	0.43
	20	(0.18)	50	(0.40)	125	(0.43)
9–10	0.57	0.32	0.60	0.34	0.65	0.40
	17	(0.45)	65	(0.32)	117	(0.34)
11–13	0.70	0.32	0.57	0.45	0.66	0.38
	19	(0.22)	60	(0.37)	142	(0.58)
14–17	0.59	0.44	0.64	0.38	0.62	0.33
	13	(0.64)	75	(0.48)	146	(0.28)
18–22	0.75	0.38	0.61	0.27	0.67	0.34
	15	(0.30)	75	(0.22)	126	(0.28)
23–31	0.63	0.96	0.55	0.29	0.61	0.33
	17	(0.84)	62	(0.24)	123	(0.27)
Greater than 32	0.33	0.24	0.57	0.30	0.47	0.25
	7	(0.14)	54	(0.28)	87	(0.24)

Firms are classified by the concentration of the credit markets where the firm is located and by the age of the firm. Firm age is years since acquired by the present owners. The SBA survey classifies the Metropolitan Statistical Area or county where the firm is headquartered into three categories on the basis of the commercial bank deposit Herfindahl index of the area. The most competitive market has a Herfindahl index of less than 0.10, and the most concentrated market has a Herfindahl index of more than 0.18. The debt-to-asset ratio includes only debt from institutions in the numerator. Loans from owners and family are not included.

TABLE III
TRADE CREDIT USAGE ACROSS CREDIT MARKETS WITH DIFFERING DEGREES OF COMPETITION

	Most competitive market	Middle market	Most concentrated market
	Median (Number of obs.)	Median (Number of obs.)	Median (Number of obs.)
Fraction of purchases offered on credit	0.90 (231)	0.90 (818)	0.90 (1597)
Days payable outstanding (DPO)	15.83 (160)	16.22 (594)	14.58 (1232)
Fraction of firms taking less than 10 percent of the discounts	0.33	0.29	0.19
Fraction of firms taking more than 90 percent of their discounts	0.50	0.52	0.59

The most competitive market has a Herfindahl index of less than 0.10, and the most concentrated market has a Herfindahl index of more than 0.18. Days payable outstanding is $365 \times \text{Accounts Payable} / \text{Cost of Goods Sold}$.

5.3 Table III: Trade-credit usage as a constraint measure

Read:

- Raw use of trade credit is common everywhere: the median firm in all markets finances 90% of purchases on credit, and median DPO is around 15 days across the board.
- The informative margin is not access to trade credit but **how much of the discount firms forgo**. In concentrated markets, only 19% of firms take less than 10% of discounts, compared with 33% in competitive markets.
- Conversely, 59% of firms in concentrated markets take more than 90% of offered discounts, versus 50% in competitive markets.
- So firms in concentrated markets appear less forced to borrow from this expensive noninstitutional source.

5.4 Table IV: Tobit regressions for credit availability

Read:

- The dependent variable is the percentage of early-payment discounts taken, estimated with a two-sided Tobit because the observed variable is censored at 0 and 100. Fifteen percent of firms are censored at 0 and 45% at 100.
- The most concentrated-market dummy enters strongly positively: depending on specification, the coefficient is about +16.5 to +19.1 percentage points. This means firms in concentrated markets are substantially less credit constrained.
- Relationship strength matters. The log length of the longest relationship has a coefficient around +18.4 to +19.7, while borrowing from more institutions reduces discounts taken by about 11 percentage points.
- The main concentration result survives controls for rural location, sales growth, and unit banking laws, which strengthens the causal interpretation.

Independent variable	I	II	III	IV
Firm characteristics				
Log(book value of assets)	7.21 ¹ (1.70)	7.26 ¹ (1.70)	7.10 ¹ (1.79)	7.13 ¹ (1.79)
Operating profits/Book assets	5.88 ⁵ (2.79)	5.95 ⁵ (2.80)	5.22 ¹⁰ (2.82)	5.35 ¹⁰ (2.84)
Firm is a corporation (0,1)	-8.96 (5.88)	-9.10 (5.89)	-10.84 ¹⁰ (6.24)	-10.54 ¹⁰ (6.24)
Log firm age (in years)	8.69 ¹ (3.35)	8.74 ¹ (3.35)	10.19 ¹ (3.73)	10.39 ¹ (3.73)
Firm is located in an MSA	-7.44 (5.45)	-6.93 (5.47)	-4.32 (5.75)	-4.97 (5.78)
Sales growth (1986–1987)			4.40 (3.86)	4.41 (3.86)
Firm is located in a unit banking state		9.36 (9.94)		
Relationship characteristics				
Log(length in years of longest relationship)	18.64 ¹ (3.84)	18.42 ¹ (3.85)	19.70 ¹ (4.05)	19.64 ¹ (4.05)
Debt from financial service provider (percent)	2.54 (5.74)	2.11 (5.75)	3.37 (6.03)	3.51 (6.03)
Number of institutions from which firm borrows	-11.62 ¹ (3.16)	-11.56 ¹ (3.16)	-11.40 ¹ (3.33)	-11.35 ¹ (3.33)
Financial market concentration				
Herfindahl > 0.18 (0,1)	16.54 ¹ (5.66)	17.21 ¹ (5.70)	16.87 ¹ (5.98)	19.05 ¹ (6.30)
Herfindahl < 0.10 (0,1)				11.48 (10.35)
Number of observations	1459	1459	1339	1339
–Log likelihood	4120.1	4119.7	3743.0	3742.4

The dependent variable is the percentage of offered early payment discounts taken by the firm. The coefficient estimates are from a tobit regression with two-sided censoring. Fifteen percent of the observations are censored at 0 percent; 45 percent of the observations are censored at 100 percent. Standard errors are in parentheses. The regression also includes seven industry dummies, three region dummies, and a constant.

¹Coefficient is significant at the 1 percent level.

⁵Coefficient is significant at the 5 percent level.

¹⁰Coefficient is significant at the 10 percent level.

Firm characteristics	Most competitive	Middle market	Most concentrated
Firm age < 10 years	53.4 ⁵	54.7	63.0
	50.0	60.0	90.0
	(44.6)	(44.6)	(41.5)
Firm age > 10 years	93	290	593
	68.4 ¹⁰	67.2	76.6
	100.0	95.0	100.0
	(41.7)	(41.5)	(35.9)
	69	331	640

We tested the equality of the mean percent of early payment discounts taken in the most concentrated and most competitive markets. The test is done separately for young (firms less than or equal to ten years old) and old firms (firms greater than ten years old). Differences in the means are noted for significance levels of 10 percent (¹⁰), 5 percent (⁵), and 1 percent (¹).

5.5 Table V: Age splits in the trade-credit measure

Read:

- Among young firms, average discounts taken rise from 53.45 in the most competitive markets to 63.0 in the most concentrated markets.

TABLE VI
BORROWING COSTS ACROSS MARKET STRUCTURES

	Young firms (Age $\leq$ 10 years)	Old firms (Age $>$ 10 years)
Most competitive market (Herfindahl $<$ 0.10)	11.50 (2.31)	10.64 (2.19)
	59	48
Most concentrated market (Herfindahl $>$ 0.18)	11.38 (2.34)	11.04 (2.36)
	477	403

The SBA survey classifies the Metropolitan Statistical Area or county where the firm is headquartered into three categories on the basis of the commercial bank deposit Herfindahl index of the area. The most competitive markets are those with a value of the Herfindahl index of less than 0.10. The most concentrated markets are those with a Herfindahl index of more than 0.18. Young firms are ten years old (the median age) or less. Old firms are more than ten years old. We report the mean interest rate, the standard deviation, and the number of observations in each cell.

- The median young firm takes 50% of discounts in the most competitive markets, 60% in the middle market, and 90% in the most concentrated market.
- For old firms, these cross-market differences are much weaker: the median is essentially 100% in both the competitive and concentrated markets.
- This table is consistent with the idea that market structure matters most when firms are young and informationally opaque, though the interaction is not always statistically sharp in this reduced-form split.

5.6 Table VI: Raw borrowing costs

Read:

- The unconditional means already show the life-cycle cross-over predicted by the model.
- In the most competitive markets, young firms pay 11.50% while old firms pay 10.64%.
- In the most concentrated markets, young firms pay slightly less, 11.38%, but old firms pay more, 11.04%.
- So concentrated markets appear to offer slightly cheaper credit early and slightly more expensive credit later.

5.7 Table VII and Figure II: Interest-rate regressions and the age profile of loan rates

Read:

- After controlling for loan characteristics, firm characteristics, macro interest-rate conditions, relationship variables, industry, and region, firms in concentrated markets start out with lower borrowing costs. The concentrated-market intercept is around -1.29 percentage points relative to the competitive benchmark for the youngest firms.
- The key result is the age slope. In the most competitive market, the coefficient on log age is about -0.725 ; in the most concentrated market it is only about -0.154 .
- The implied decline in rates over the first ten years is about 167 basis points in competitive markets but only about 36 basis points in concentrated markets.

TABLE VII
THE EVOLUTION OF BORROWING COSTS ACROSS MARKET STRUCTURES

Independent variable	I	II	III
Credit market conditions			
Herfindahl greater than 0.18 (0,1)	-1.285 ¹⁰ (0.686)		-1.347 ¹⁰ (0.690)
Herfindahl between 0.10 and 0.18 (0,1)	-1.704 ⁵ (0.721)		-1.759 ⁵ (0.724)
Log of firm age if Herfindahl is greater than 0.18	-0.154 ¹⁰ (0.088)		-0.153 ¹⁰ (0.088)
Log of firm age if Herfindahl is between 0.10 and 0.18	-0.135 (0.127)		-0.134 (0.127)
Log of firm age if Herfindahl is less than 0.10	-0.725 ¹ (0.273)		-0.732 ¹ (0.273)
Log of firm age if firm is in an MSA		-0.240 ⁵ (0.112)	
Log of firm age if firm is not in an MSA		-0.154 ¹⁰ (0.090)	
Interest rate variables			
Floating rate loan (0,1)	-0.477 ⁵ (0.187)	-0.459 ⁵ (0.188)	-0.479 ⁵ (0.187)
Prime rate	0.278 ¹ (0.032)	0.272 ¹ (0.032)	0.277 ¹ (0.032)
Term structure spread	0.011 (0.084)	0.014 (0.085)	0.010 (0.084)
Default spread	0.357 ⁵ (0.153)	0.364 ⁵ (0.153)	0.352 ⁵ (0.153)
Firm characteristics			
Log (book value of assets)	-0.295 ¹ (0.048)	-0.301 ¹ (0.048)	-0.295 ¹ (0.048)
Total debt/book assets	-0.011 (0.147)	-0.024 (0.148)	-0.003 (0.148)
Firm is a corporation	-0.158 (0.145)	-0.173 (0.145)	-0.159 (0.145)
Number of banks from which firm borrows	0.379 ¹ (0.088)	0.378 ¹ (0.088)	0.377 ¹ (0.088)
Firm is in an MSA	-0.021 (0.138)	0.074 (0.343)	-0.031 (0.139)
Firm is in a unit banking state			-0.213 (0.244)
Loan is from a bank (0,1)	0.100 (0.193)	0.140 (0.192)	0.107 (0.193)
Loan is from a nonfinancial firm (0,1)	-1.148 ¹ (0.367)	-1.140 ¹ (0.367)	-1.144 ¹ (0.367)
Number of observations	1277	1277	1277
Adjusted R ²	0.160	0.154	0.159

The rate quoted on the firm's most recent loan is the dependent variable. The regression also includes seven industry dummies, three regional dummies, six dummy variables for the type of assets with which the loan is collateralized, and an intercept. The prime rate is the rate at the time the loan was made. The term structure spread is the difference between the ten-year government bond yield and the three-month T-bill yield at the time the loan was made. The default spread is the difference between the BAA corporate bond yield and the ten-year government bond yield at the time the loan was made.

¹ Coefficient is significant at the 1 percent level.

⁵ Coefficient is significant at the 5 percent level.

¹⁰ Coefficient is significant at the 10 percent level.

Loan Rates Across Market Structures

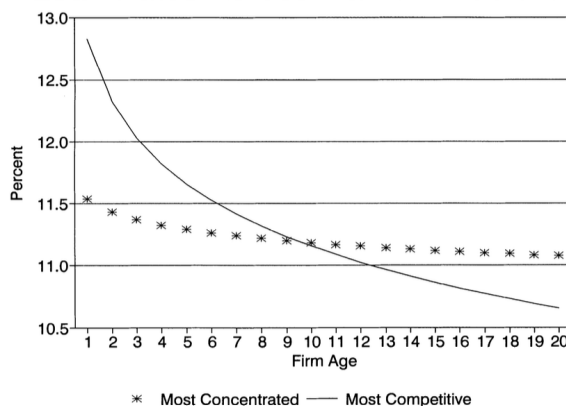


FIGURE II

The loan rates are based on estimates from Table VII. All variables, except the firm's age and the market type (most competitive versus most concentrated) were set equal to the sample means. The average loan rate in this sample is 11.1 percent.

- Figure II visualizes the theoretical mechanism directly: rates start lower in concentrated markets but do not fall nearly as much with firm age. Controls for MSA status or unit banking laws do not overturn this pattern.

6 Short summary

- **Conclusion.** Young firms obtain more institutional finance in concentrated credit markets, appear less credit constrained, and face borrowing costs that are smoothed more heavily over time.
- **Identification insight.** The evidence is persuasive because the paper shows that concentrated markets do not simply contain better firms, and because the cross-market pattern appears both in actual institutional borrowing and in an alternative rationing measure based on trade-credit discounts.
- **Economic lesson.** Credit-market competition can reduce the value of lending relationships by preventing lenders from subsidizing young firms up front and recovering rents later.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes a concrete mechanism through which credit-market concentration can increase the availability of finance to young and opaque firms.
- It also shows that this is not a free lunch: concentrated markets appear to transfer surplus intertemporally, helping borrowers when young but charging them more later.
- The empirical evidence lines up closely with the model's qualitative predictions on access, constraint measures, and the life-cycle path of loan rates.

7.2 Economic intuition

The core intuition is simple. A young firm's current cash flow may be low even though its future prospects are good. If the lender cannot expect to capture any of that future upside, it must charge a high rate today, and that rate may distort the firm's incentives so badly that it will not lend at all. By contrast, a lender with market power can effectively take an implicit stake in the future relationship: it lends more cheaply at the start, expecting to earn rents later once the firm's quality is known and the relationship is established. This dynamic makes concentrated markets better at financing young firms, but also makes mature borrowers pay a long-run cost for that flexibility.

7.3 Takeaway

This paper should be remembered as a classic statement of the **availability-versus-price trade-off** in credit markets. More competition lowers long-run borrowing costs for established firms, but it can also undermine relationship lending and make it harder for young firms to get financed in the

first place. The paper does not claim that concentration is always welfare improving. Instead, it shows that relationship finance relies on some mechanism for intertemporal surplus sharing, and that competition can weaken exactly that mechanism.